MONISHA R S

GRADUATE SOFTWARE ENGINEER | PYTHON • JAVA • SQL

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PROFESSIONAL SUMMARY

Artificial Intelligence and Data Science graduate with 8.32/10 CGPA and hands-on experience developing robust software solutions using Python, Java, and SQL. Possesses a strong foundation in Data Structures and Algorithms, Object-Oriented Programming (OOP), DBMS, analytical thinking, and core software engineering principles. Experienced in building end-to-end applications spanning software development, data processing, automation, REST APIs, and database integration. Fast learner adept at team collaboration, technical communication, and delivering reliable project outcomes within tight timelines.

TECHNICAL SKILLS

Languages:	Python, Java, SQL
Core CS Concepts:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, RDBMS
Software Engineering:	Application Development, REST APIs, Data Processing, Automation, Debugging, Software Testing
Databases & Tools:	SQL, SQLite, MongoDB, Git, GitHub, Postman, Streamlit
AI & Data Science:	Machine Learning, NLP, Generative AI, RAG, LLMs, Pandas, NumPy, Scikit-learn, OpenCV, Pytesseract, LangChain
Cloud & APIs:	Google Gemini API, Google Cloud Vertex AI

TECHNICAL PROJECTS

Agentic AI-Driven Decision Support Platform for Startups & MSMEs — Python | LangChain | Gemini API

- Developed an end-to-end Python-based software solution for business-profile analysis and automated government scheme discovery.
- Implemented automated workflows for structured data processing, document feature extraction, and eligibility analysis.
- Integrated semantic retrieval, contextual reasoning, and business logic to seamlessly process structured and unstructured information.
- Designed modular, scalable software workflows that reduced manual scheme-search operational effort by 80%+.

Retrieval-Augmented Generation (RAG) PDF Question Answering Chatbot — Python | ChromaDB | Streamlit

- Built a Python application for efficient semantic processing and querying across complex, multi-page PDF documents.
- Implemented vector document indexing and context-aware retrieval using ChromaDB, Google Gemini API, and LangChain.
- Designed an interactive, user-friendly Streamlit web interface for seamless document upload and real-time QA.
- Optimized vector index retrieval pipelines to consistently achieve sub-2-second query response latency.

Automated ID Card Scanner & Extraction System — Python | OpenCV | Pytesseract | MongoDB

- Developed an OCR-based document processing system utilizing Python, OpenCV, and Pytesseract for automated identity extraction.
- Implemented image preprocessing routines including orientation correction, adaptive thresholding, and contrast enhancement.
- Boosted textual extraction accuracy by 25% through specialized image pre-filtering and OCR parameter tuning.
- Integrated structured JSON formatting and MongoDB database storage, reducing manual data-entry processing time by 75%.

Automated NLP Video Translator & Subtitle Generator Python | Speech-to-Text | Translation APIs

- Engineered an automated NLP software pipeline for high-accuracy speech-to-text transcription and regional language translation.
- Implemented audio processing and sequence translation workflows for automated, time-synced subtitle generation.
- Designed robust data pipelines converting unstructured video/audio inputs into structured multilingual output.

EDUCATION

Sona College of Technology 2022 – 2026

Bachelor of Technology (B.Tech) – Artificial Intelligence and Data Science **CGPA: 8.32** / **10.0** (Equivalent GPA: 3.33 / 4.00)

Sri Vijay Vidyalaya Matric Higher Secondary School Tamil Nadu

*Higher Secondary Certificate (HSC): **87.16%** • Secondary School Leaving Certificate (SSLC): **98.60%***

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** Prompt Design in Vertex AI (Google Cloud), SQL Intermediate (HackerRank), Introduction to MongoDB (MongoDB University), Introduction to Java (NPTEL), Introduction to Internet of Things (NPTEL).
- **Hackathon Achievement:** Top 50 Rank in Thryve Digital National Healthcare Hackathon out of 5,000+ national entries.
- **Competitions:** Runner-Up at Connextion (MIT Samhita) | Top 40 Rank at Xyntra (Rajalakshmi Engineering College).
- **Academic Recognition:** Recipient of Academic Appreciation Award at Sona College of Technology (2023–2026).

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **PORT'25 Department Symposium Organizer:** Coordinated technical tracks, speaker logistics, and project exhibitions for 300+ attendees.
- **IE&EQ Club Organizer:** Conducted technical coding workshops, peer mentoring, and hands-on technology learning sessions for 100+ junior students.
- **NSS Volunteer & Organizer:** Led community outreach initiatives and social welfare campaigns.